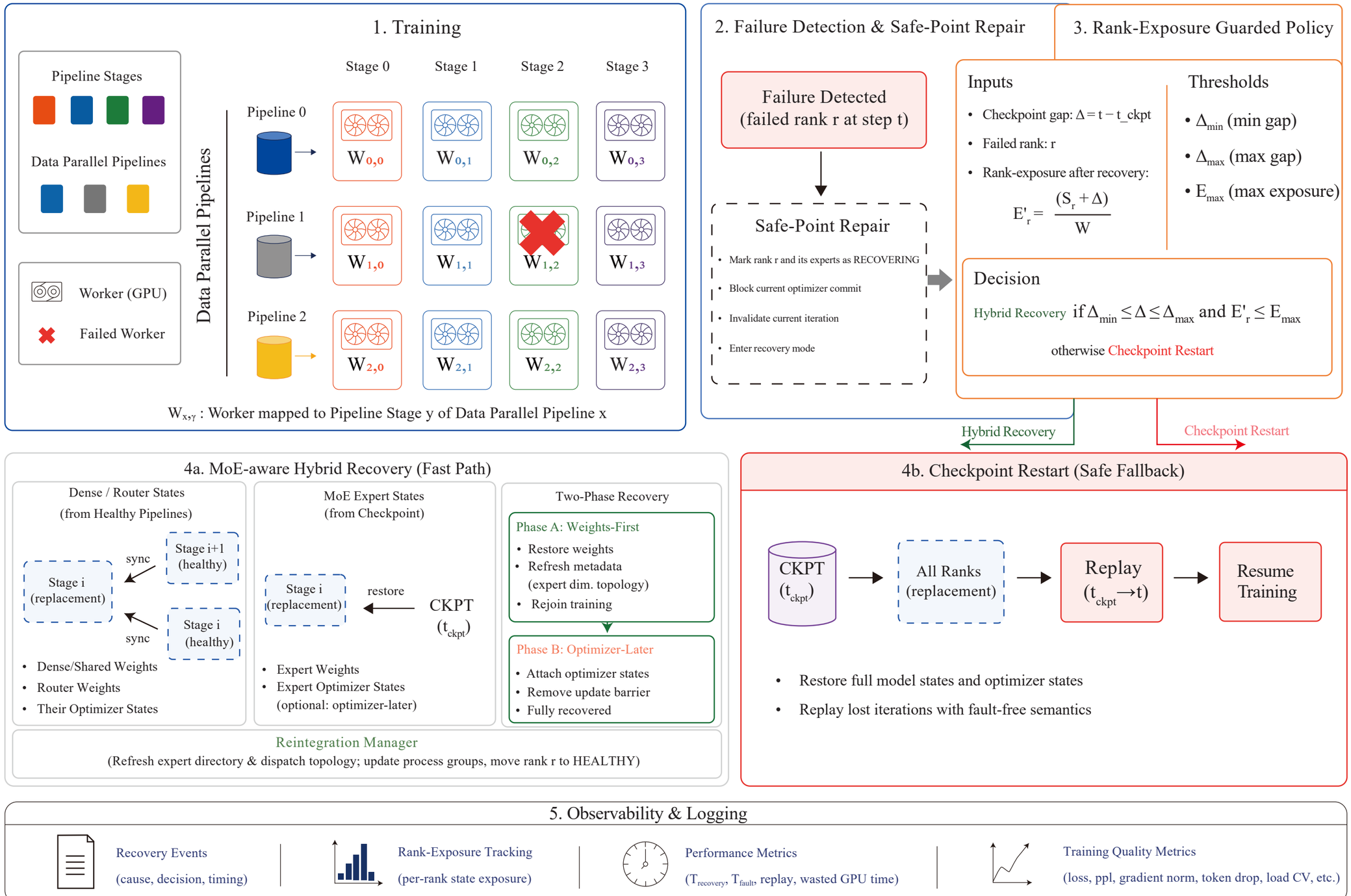

1. Training
Pipeline Stages
Data Parallel Pipelines
Worker (GPU)
Failed Worker
Data Parallel Pipelines
Stage 0
Stage 1
Stage 2
Stage 3
Pipeline 0
Pipeline 1
Pipeline 2
W0,0
W0,1
W0,2
W0,3
W1,0
W1,1
W1,2
W1,3
W2,0
W2,1
W2,2
W2,3
Wx,y : Worker mapped to Pipeline Stage y of Data Parallel Pipeline x
2. Failure Detection & Safe-Point Repair
Failure Detected
(failed rank r at step t)
Safe-Point Repair
Mark rank r and its experts as RECOVERING
Block current optimizer commit
Invalidate current iteration
Enter recovery mode
3. Rank-Exposure Guarded Policy
Inputs
Checkpoint gap: Δ = t − t_ckpt
Failed rank: r
Rank-exposure after recovery:
E'r = (Sr + Δ) / W
Thresholds
Δmin (min gap)
Δmax (max gap)
Emax (max exposure)
Decision
Hybrid Recovery if Δmin ≤ Δ ≤ Δmax and E'r ≤ Emax
otherwise Checkpoint Restart
Hybrid Recovery
Checkpoint Restart
4a. MoE-aware Hybrid Recovery (Fast Path)
Dense / Router States
(from Healthy Pipelines)
Stage i+1
(healthy)
sync
Stage i
(replacement)
sync
Stage i
(healthy)
Dense/Shared Weights
Router Weights
Their Optimizer States
MoE Expert States
(from Checkpoint)
Stage i
(replacement)
restore
CKPT
(tckpt)
Expert Weights
Expert Optimizer States
(optional: optimizer-later)
Two-Phase Recovery
Phase A: Weights-First
Restore weights
Refresh metadata
(expert dim. topology)
Rejoin training
Phase B: Optimizer-Later
Attach optimizer states
Remove update barrier
Fully recovered
Reintegration Manager
(Refresh expert directory & dispatch topology; update process groups, move rank r to HEALTHY)
4b. Checkpoint Restart (Safe Fallback)
CKPT
(tckpt)
All Ranks
(replacement)
Replay
(tckpt→t)
Resume
Training
Restore full model states and optimizer states
Replay lost iterations with fault-free semantics
5. Observability & Logging
Recovery Events
(cause, decision, timing)
Rank-Exposure Tracking
(per-rank state exposure)
Performance Metrics
(Trecovery, Tfault, replay, wasted GPU time)
Training Quality Metrics
(loss, ppl, gradient norm, token drop, load CV, etc.)